



Signals and Systems

Discrete Time Fourier Series

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9. Convolution property in time domain

If $x_1(n) \xleftrightarrow{F} X_1(\omega)$ and $x_2(n) \xleftrightarrow{F} X_2(\omega)$

then

$$x_1(n) * x_2(n) \xleftrightarrow{F} X_1(\omega) X_2(\omega)$$

Proof:
$$x_1(n) * x_2(n) = \sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k)$$

$$F[x_1(n) * x_2(n)] = \sum_{n=-\infty}^{\infty} x_1(n) * x_2(n) e^{-j\omega n}$$

$$F[x_1(n) * x_2(n)] = \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k) \right] e^{-j\omega n}$$

$$F[x_1(n) * x_2(n)] = \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k) \right] e^{-j\omega n}$$

Interchanging the order of summation

$$F[x_1(n) * x_2(n)] = \sum_{k=-\infty}^{\infty} x_1(k) \sum_{n=-\infty}^{\infty} x_2(n-k) e^{-j\omega n}$$

$$F[x_1(n) * x_2(n)] = \sum_{k=-\infty}^{\infty} x_1(k) F[x_2(n-k)]$$

$$= \sum_{k=-\infty}^{\infty} x_1(k) X_2(\omega) e^{-j\omega k} = X_2(\omega) \sum_{k=-\infty}^{\infty} x_1(k) e^{-j\omega k}$$

$$= X_2(\omega) X_1(\omega)$$

hence

$$x_1(n) * x_2(n) \xleftrightarrow{F} X_1(\omega) X_2(\omega)$$

10. Convolution property in frequency domain

If $x_1(n) \xleftrightarrow{F} X_1(\omega)$ and $x_2(n) \xleftrightarrow{F} X_2(\omega)$

then

$$x_1(n)x_2(n) \xleftrightarrow{F} X_1(\omega) * X_2(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda)X_2(\omega - \lambda)d\lambda$$

Proof:

$$\begin{aligned} F[x_1(n)x_2(n)] &= \sum_{n=-\infty}^{\infty} x_1(n)x_2(n)e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda)e^{j\lambda n} d\lambda \right] x_2(n)e^{-j\omega n} \end{aligned}$$

Interchanging the order of summation and integration

$$F[x_1(n)x_2(n)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda) \left[\sum_{n=-\infty}^{\infty} x_2(n)e^{j\lambda n} e^{-j\omega n} \right] d\lambda$$

$$\begin{aligned}
F[x_1(n)x_2(n)] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda) \left[\sum_{n=-\infty}^{\infty} x_2(n) e^{j\lambda n} e^{-j\omega n} \right] d\lambda \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda) \left[\sum_{n=-\infty}^{\infty} x_2(n) e^{-j(\omega-\lambda)n} \right] d\lambda \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda) X_2(\omega - \lambda) d\lambda \\
&= X_1(\omega) * X_2(\omega)
\end{aligned}$$

hence

$$x_1(n)x_2(n) \xleftrightarrow{F} X_1(\omega) * X_2(\omega)$$

11. Parseval's Theorem

If $x(n) \xleftrightarrow{F} X(\omega)$
then

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$$

Proof:

$$\begin{aligned} \sum_{n=-\infty}^{\infty} |x(n)|^2 &= \sum_{n=-\infty}^{\infty} x(n) x^*(n) \\ &= \sum_{n=-\infty}^{\infty} x(n) \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega \right]^* \\ &= \sum_{n=-\infty}^{\infty} x(n) \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) e^{-j\omega n} d\omega \right] \end{aligned}$$

Interchanging the order of summation and integration

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) \left[\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right] d\omega$$

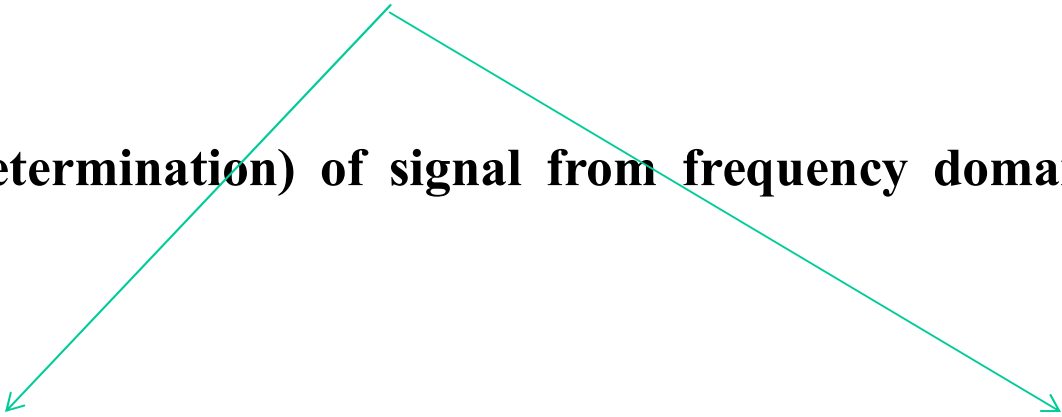
$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) X(\omega) d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$$

hence

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$$

Inverse Fourier Transform



Recovering (determination) of signal from frequency domain into time domain (x(n)).

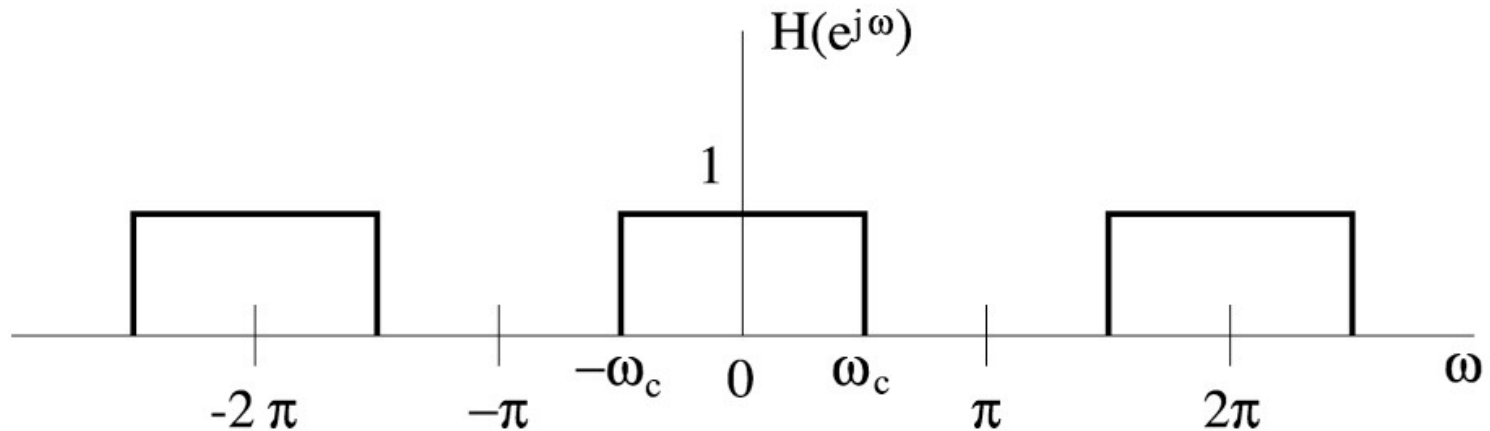
1. Using Inverse Fourier Transform Formula

$$\begin{aligned} x[n] &= F^{-1}[X(\omega)] \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega \end{aligned}$$

2. Using Partial Fraction in Standard Form

In this method, $X(\omega)$ is broken as sum of two or more whose inverse Fourier transform are known, using Partial Fraction method

Example 12: Determine the inverse Fourier transform of following spectrum



$$h[n] = F^{-1}[H(\omega)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) e^{j\omega n} d\omega$$

$$-\pi < \omega < \pi$$

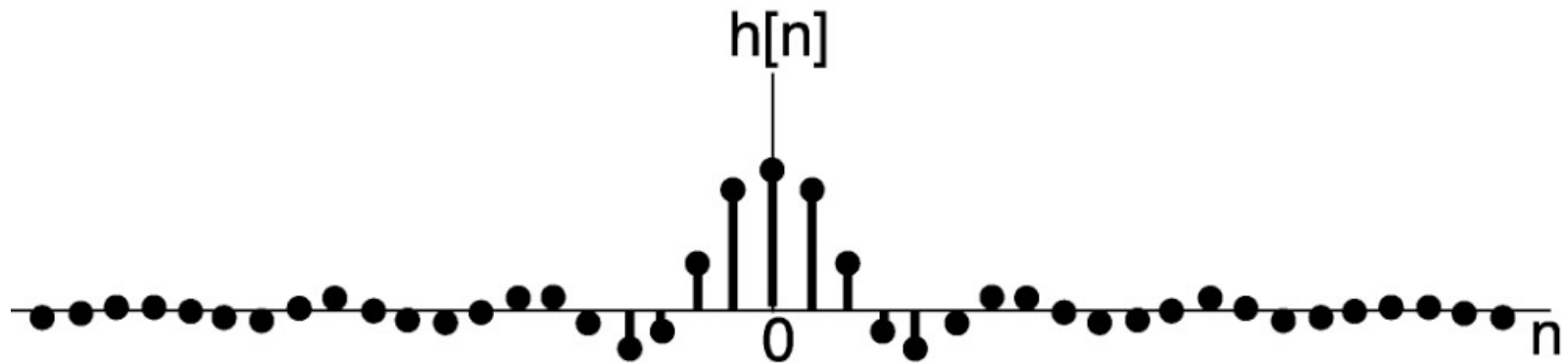
$$H(e^{j\omega}) = H(\omega) = 1, \text{ for } -\omega_c < \omega < \omega_c$$

$$= 0, \text{ otherwise}$$

$$\begin{aligned}
 h(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) e^{j\omega n} d\omega \\
 &= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} (1) e^{j\omega n} d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c} \quad n \neq 0 \\
 &= \frac{1}{(2\pi)jn} (e^{j\omega_c n} - e^{-j\omega_c n}) = \frac{\sin(\omega_c n)}{\pi n}
 \end{aligned}$$

For $n = 0$

$$h(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} (1) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} (1) d\omega = \frac{\omega_c}{\pi}$$



Example 13: Find the inverse Fourier transform of the following:

$$X(\omega) = \begin{cases} 1, & \frac{\pi}{3} \leq |\omega| \leq \frac{2\pi}{3} \\ 0, & \frac{2\pi}{3} \leq |\omega| \leq \pi \end{cases}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \left[\int_{-2\pi/3}^{-\pi/3} 1 e^{j\omega n} d\omega + \int_{\pi/3}^{2\pi/3} 1 e^{j\omega n} d\omega \right]$$

$$\begin{aligned}
&= \frac{1}{2\pi} \left[\left(\frac{e^{j\omega n}}{jn} \right)_{-2\pi/3}^{-\pi/3} + \left(\frac{e^{j\omega n}}{jn} \right)_{\pi/3}^{2\pi/3} \right] \\
&= \frac{1}{n\pi} \left[\frac{e^{-jn(\pi/3)} - e^{-jn(2\pi/3)} + e^{jn(2\pi/3)} - e^{jn(\pi/3)}}{2j} \right] \\
&= \frac{1}{n\pi} \left[\frac{e^{jn(2\pi/3)} - e^{-jn(2\pi/3)}}{2j} - \frac{e^{jn(\pi/3)} - e^{-jn(\pi/3)}}{2j} \right] = \frac{1}{n\pi} \left[\sin n \frac{2\pi}{3} - \sin n \frac{\pi}{3} \right]
\end{aligned}$$

Inverse Fourier Transform using Partial Fraction in Standard Form

$$\delta(n) \xleftrightarrow{F} 1 \qquad \delta(n - n_0) \xleftrightarrow{F} e^{-j\omega n_0}$$

$$a^n u[n] \xleftrightarrow{F} \frac{1}{1 - ae^{-j\omega}}$$

$$na^n u[n], \quad |a| < 1 \xleftrightarrow{F} \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2}$$

Example 14: Determine the inverse Fourier transform of following spectrum

$$X(\omega) = 2 + 3e^{-j\omega} + 3e^{-2j\omega} - 2.5e^{-4j\omega} - 3e^{j\omega}$$

Solution:

$$x(n) = F^{-1}[X(\omega)]$$

$$= F^{-1}[2] + 3F^{-1}[e^{-j\omega}] + 3F^{-1}[e^{-2j\omega}] - 2.5F^{-1}[e^{-4j\omega}] - 3F^{-1}[e^{j\omega}]$$

$$x(n) = 2\delta(n) + 3\delta(n-1) + 3\delta(n-2) - 2.5\delta(n-4) - 3\delta(n+1)$$

$$x(n) = \{-3, 2, 3, 3, 0, -2.5\}$$


Example 15: Determine the inverse Fourier transform of following spectrum

$$X(\omega) = \frac{2}{1 - \frac{1}{2}e^{-j\omega}} - \frac{1.5}{1 - \frac{1}{3}e^{-j\omega}} + \frac{2.5e^{-j\omega}}{(1 - \frac{1}{5}e^{-j\omega})^2}$$

Solution:

$$x(n) = F^{-1}\left(\frac{2}{1 - \frac{1}{2}e^{-j\omega}}\right) - F^{-1}\left(\frac{1.5}{1 - \frac{1}{3}e^{-j\omega}}\right) + F^{-1}\left(\frac{2.5e^{-j\omega}}{(1 - \frac{1}{5}e^{-j\omega})^2}\right)$$

$$x(n) = 2\left(\frac{1}{2}\right)^n u(n) - 1.5\left(\frac{1}{3}\right)^n u(n) + 2.5n\left(\frac{1}{5}\right)^n u(n)$$

Example 16: Determine the inverse Fourier transform of following spectrum

$$X(\omega) = \frac{2}{1 - \frac{1}{2}e^{-j\omega}} - \frac{1.5e^{-2j\omega}}{1 - \frac{1}{3}e^{-j\omega}}$$

Solution:

$$x(n) = F^{-1}\left(\frac{2}{1 - \frac{1}{2}e^{-j\omega}}\right) - F^{-1}\left(\frac{1.5e^{-2j\omega}}{1 - \frac{1}{3}e^{-j\omega}}\right)$$

$$x(n) = 2\left(\frac{1}{2}\right)^n u(n) - 1.5\left(\frac{1}{3}\right)^{n-2} u(n-2)$$

Using time shifting prop.